

Multiple Regression I: Conditioning, Checks, and Coordinated Questions

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i Chapter overview

One study can owe several answers. This chapter keeps the occupation records of Chapter 7 and asks three questions of the 98 with a recorded type: how income varies with education across them, how it differs with education at the same prestige, percentage of women, and type, and whether a fuller learning rule predicts held-out occupations better than the education-only rule. We develop what a partial slope compares and why it can differ from the marginal slope, estimate it with a stated working model, check support, form, influence, and precision, and calculate a fitted profile, its two intervals, and a standardized quantity. We then answer the prediction question with shared held-out folds, read the three reported answers and their reasons before evaluating each and their synthesis. By the end of the chapter you will be able to fit and read an additive multiple regression in R, keep changes of records, conditioning, coordinates, and uncertainty apart, judge a held-out comparison, and evaluate coordinated answers without merging them.

A report may ask whether education and income move together, whether that relationship remains when other occupational characteristics are held constant, and whether those characteristics help predict income. These questions can inform one another, but answering one does not discharge the other two obligations.

We use a **chapter-authored retrospective teaching study**, not a published study with a newly discovered analysis plan. Its records are the *Prestige* data distributed with the *carData* package (Fox et al., 2022; Fox & Weisberg, 2019). The analyses and short report below are our teaching record; the package documentation supplies the data descriptions, not the claims written for this chapter.

A rounded coefficient equation explains a comparison; evaluate the unrounded fitted model when checking its last digits. Each visible R example follows from the earlier visible preparation in this chapter; Chapter 7 supplies familiarity, not an existing R session. Chapter 5 held one variable fixed by residualizing: its partial correlation removed that variable's fitted linear component from two others before correlating what remained. A fitted equation with several predictors holds the others fixed directly, because two of its fitted means that differ in one predictor share every other term. We use that idea with several controls at once, track the full model's uncertainty, and distinguish a conditional comparison from an average over named profiles. Chapter 9, Section 2, applies the same idea to adjusted group means in the analysis of covariance.

1 Three Questions in the Same Occupation Records

1.1 One study can owe several answers

Table 1

Three separately owed answers in the occupation teaching study.

Pair and role	Question	Estimand and owed answer
1. Contextual description	How does income vary linearly with education across the 98 retained occupations?	The finite-record marginal least-squares slope; a descriptive answer for these records
2. Primary conditional association	How does income differ with education at the same prestige, percentage of women, and type, within support?	The linear conditional-mean coefficient β_E under the declared working model; fitted b_E is its estimate from the 98 retained records
3. Secondary prediction analysis	Does the full additive learning rule predict eligible same-source occupation records with lower expected absolute error than the education-only rule?	Difference in expected absolute error of the two learning procedures at the declared training sizes; an internal validation answer

These are three question–answer pairs, each with **one argument with linked layers**. They are not three statistical strands of one argument: the report owes separate conclusions about marginal association, conditional association, and prediction. If several estimands were jointly necessary for *one* integrated conclusion, they could instead be parallel statistical strands. Merely printing several coefficients or diagnostics does not create either several questions or several strands.

1.2 Keep the records, time, and units visible

A row represents an **occupation**, not an individual worker. Income is the occupation’s average annual income in 1971 Canadian dollars; we divide it by 1,000. Education is average years of education, and *women* is the percentage of women in the occupation. Prestige ratings were collected in a mid-1960s survey, whereas the income and education summaries come from 1971. Occupational type is blue-collar (*bc*), professional/managerial/ technical (*prof*), or white-collar (*wc*). These historical and aggregate measurements are not interchangeable with current individual earnings.

The earlier simple-regression analysis used all 102 occupations with education and income recorded. The declared full variable set leaves 98 complete records: athletes, newsboys, babysit-

ters, and farmers have missing type. The retained type counts are 44, 31, and 23 respectively. We keep the original row order, which will also make the validation split reproducible.

```
all_occ <- carData::Prestige
all_occ$occupation <- rownames(all_occ)
all_occ$income_k <- all_occ$income / 1000
all_occ$type <- factor(all_occ$type, levels = c("bc", "prof", "wc"))
vars <- c("income_k", "education", "prestige", "women", "type")
d <- all_occ[complete.cases(all_occ[, vars]), ]
c(original = nrow(all_occ), retained = nrow(d))
```

```
original retained
      102       98
```

```
table(d$type)
```

```
bc prof  wc
44  31  23
```

We refit the education-only model on these **same 98 records** before comparing adjustment specifications. Moving from 102 to 98 changes which records are described. It must not be presented as an adjustment effect. Complete-case selection makes the computations comparable; it does not establish that the omitted occupations are irrelevant or that these records are a probability sample of Canadian occupations.

1.3 Declare the model question before reporting an interval

Pair 1 is finite-record description. Its slope is exactly calculable once the records and least-squares rule are specified.

Pair 2 has one intended estimand: the education coefficient β_E in the declared linear conditional-mean working model. The fitted coefficient b_E is its estimate from the 98 retained records. Although b_E is exactly calculable once the rows and fitting rule are fixed, that fact does not create a second finite-record estimand. A confidence interval additionally posits that occupation outcomes could vary independently around the stated linear conditional mean at the recorded predictors. This is not a claim that the dataset came from a verified random sample or that income was randomly assigned.

For the classical interval we additionally assume a common error variance; normal errors make its finite-sample t reference exact. Robust uncertainty relaxes the common-variance requirement

under its own large-sample conditions. It does not create independence or linearity. We will check how credible and stable the working-model reading is before judging its answer.

A **best linear projection** can instead summarize a curved conditional relationship under a specified distribution of predictors. That is a different interpretation, not a rescue silently adopted after discovering curvature. The main sections use the stated linear conditional-mean interpretation; Chapter 9, Appendix A, develops the alternative and its limitations.

Pair 3 will declare a different hypothetical repetition: train a rule on eligible, exchangeable same-source records, then predict a new record without using its outcome in training. Its justification cannot be supplied by the coefficient interval in Pair 2.

2 From Marginal to Conditional Comparisons

2.1 What a partial slope compares

Chapter 5's partial correlation held one variable fixed by residualizing both variables on it. A fitted equation holds its other predictors fixed directly: specify what changes, keep the other predictor values the same, and subtract the two fitted means.

Let e be education, p prestige, w percentage of women, and t type. The fitted additive equation is

$$\widehat{m}(e, p, w, t) = b_0 + b_E e + b_P p + b_W w + b_{\text{prof}} D_{\text{prof}} + b_{\text{wc}} D_{\text{wc}}.$$

An indicator D is 1 for its named category and 0 otherwise. Blue-collar is the reference type, with both indicators zero. This model has six coefficients, including its intercept.

Take two profiles with the same p, w, t and education differing by one year. Subtract their fitted means:

$$\begin{aligned} \widehat{m}(e + 1, p, w, t) - \widehat{m}(e, p, w, t) \\ = b_E(e + 1) - b_E e = b_E. \end{aligned}$$

All the common terms cancel. Thus b_E compares modeled occupation-level income at the **same prestige, percentage of women, and type**. A two-year difference corresponds to $2b_E$ under this additive model.

“Holding constant” describes this mathematical comparison. It does not say that education was changed experimentally, that two observed occupations match exactly on every other characteristic, or that unmeasured differences are controlled. An additive slope also assumes the same education comparison across the included predictor values. Allowing it to vary with another predictor is the interaction question taken up later.

2.2 Why the marginal slope can differ

A short known-process example makes the distinction visible. Suppose

$$Y = 50 + 2X + 5Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0,$$

with finite variances and $\text{Cov}(X, Z)/\text{Var}(X) = 0.65$. The conditional X coefficient is 2. The marginal linear slope is

$$\frac{\text{Cov}(X, Y)}{\text{Var}(X)} = 2 + 5 \frac{\text{Cov}(X, Z)}{\text{Var}(X)} = 2 + 5(0.65) = 5.25.$$

Figure 1 draws one generated sample of 250 cases from this process, with the process's own mean lines rather than fitted ones. Each solid line holds Z at one value and rises 2 for each unit of X . The dashed marginal line rises 5.25, because the darker points, those with higher Z , sit toward the upper right: higher X comes with higher Z , and the marginal line carries that association along.

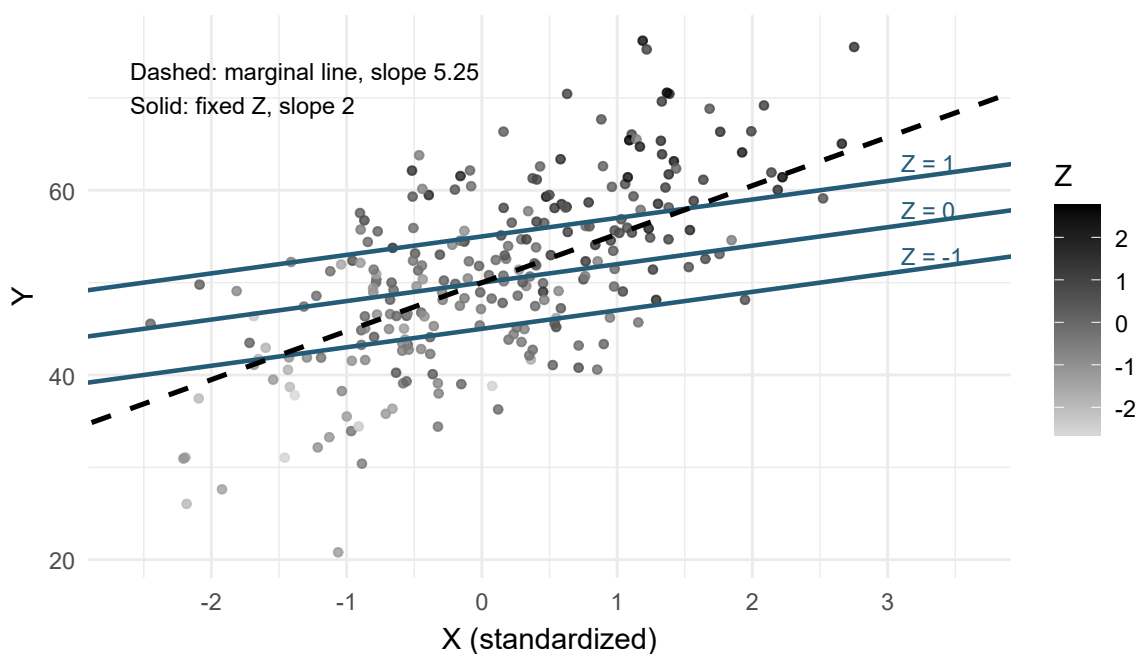


Figure 1: One generated sample from the known process, with its marginal line and three lines at fixed Z .

The conditional comparison holds Z fixed; the marginal comparison incorporates how Z moves with X in this specified process. Neither result alone says what would happen if a researcher

intervened on X . Calling 5.25 “biased” requires naming an estimand for which it is being used incorrectly; as a marginal linear summary, 5.25 is the right answer here. Appendix A retains the derivation and the repeated samples.

More generally, adding predictors can change the comparison, but it does not **always** change the scientific estimand. In a randomized study, appropriately specified covariate adjustment may estimate the same declared average assignment effect more precisely. Releveling a factor changes coefficient coordinates without changing fitted values. A different uncertainty estimator may change the interval without changing the fitted coefficient.

2.3 A small indicator bridge

The full occupation fit, displayed to four decimal places, is

$$\widehat{\text{income}}_k = 0.4205 + 0.1164e + 0.1406p - 0.0535w \\ + 0.3242D_{\text{prof}} + 0.2353D_{\text{wc}}.$$

Consider a white-collar profile with education 11, prestige 45, and women 50. Here $D_{\text{prof}} = 0$ and $D_{\text{wc}} = 1$. Using the full fitted coefficients in the substitution gives

$$\widehat{m}_* = \hat{\beta}_0 + 11\hat{\beta}_E + 45\hat{\beta}_P + 50\hat{\beta}_W + \hat{\delta}_{\text{wc}} \\ \approx 5.5886.$$

That is a fitted mean of about 5,589 dollars on the historical income scale, not a guarantee for any occupation or worker. Can these three predictor values occur together within the observed white-collar covariate cloud? Separate minimum and maximum values cannot answer that question.

Here is a checkable **joint-support witness**. A convex combination is a weighted average with nonnegative weights summing to one. The four retained white-collar records in Table 2 can be combined with the listed weights: weighting each column by them returns education 11, prestige 45 and women 50 at once.

Table 2

Four observed white-collar records recover the declared profile.

Occupation	Education	Prestige	Women	Weight
Radio/TV announcers	12.71	57.6	11.15	.2452
Tellers/cashiers	10.64	42.3	91.76	.2310
Receptionists	11.04	38.7	92.86	.2261
Sales supervisors	9.84	41.5	17.04	.2976

The weights are all positive and sum to one; the table rounds them for reading, and the Computer Lab checks the weighted averages with the full-precision weights. The witness establishes membership in the joint covariate hull **within white-collar type**, using all three continuous predictors in the full model. It does not show that many occupations lie near this profile, that observed occupations match exactly, or that a causal comparison is supported. Local data may still be sparse.

The intercept refers to zero education, prestige, and women in the reference type. It helps construct the fitted surface; it need not describe a plausible occupation. Likewise, one type coefficient is not “the effect of occupational type” as a whole. In Chapter 6’s indicator fit the factor stood alone, so each coefficient was a difference between group means; here each type coefficient compares fitted means at the same education, prestige, and women. Appendix A retains the group-mean and releveling calculation; Chapter 9 develops contrasts, block tests, and the analysis-of-variance identity.

3 Estimate the Conditional Association

3.1 Compare specifications on the same rows

```
m0 <- lm(income_k ~ education, data = d)
m1 <- lm(income_k ~ education + prestige, data = d)
m2 <- lm(income_k ~ education + prestige + women + type, data = d)
coef(m2)
```

```
(Intercept)  education    prestige      women  typeprof
  0.4205235    0.1164336    0.1406054  -0.0535047  0.3241934
      typewc
  0.2353306
```

Table 3

The model sequence separates sample composition from adjustment.

Records and specification	Education		Role here
	slope	In-sample R^2	
Earlier 102, education only	0.8988	.3336	Earlier result; not the adjustment baseline
Same 98, education only	0.8830	.3296	Pair 1’s marginal description
Same 98, education and prestige	−0.2179	.4996	Intermediate comparison
Same 98, full declared predictors	0.1164	.6362	Pair 2’s estimate of β_E

Note. Slopes are thousand 1971 Canadian dollars per education year. These are fitted summaries, not causal effects. No sampling interval is attached to Pair 1’s finite-record estimand.

Education and prestige are highly correlated here, $r = .8665$. Once prestige is included, the education comparison uses variation not linearly accounted for by prestige. Adding women and type changes the conditioning set again. The movement from 0.8830 to −0.2179 to 0.1164 is not a narrative in which a true effect is successively “explained away.” It is a sequence of named comparisons in a small observational record.

The increase in R^2 describes training-data fit. Adding predictors cannot increase the least-squares residual sum of squares on the same rows when the models are nested. Better training fit does not establish better prediction on records left out of fitting.

3.2 Partial regression gives the same coefficient

Multiple regression still uses Chapter 7’s covariance-to-slope reasoning. Chapter 5 removed one control’s fitted linear component from each of two variables before correlating their residuals. Here we remove several controls. First regress education on prestige, percentage of women, and type, retaining its residuals $r_{E,i}$. Do the same for income, obtaining $r_{Y,i}$. Both auxiliary regressions include an intercept and use exactly the same 98 rows as the full fit. Then

$$b_E = \frac{\sum_i r_{E,i} r_{Y,i}}{\sum_i r_{E,i}^2} = r_{EY \cdot Z} \frac{s_{r_Y}}{s_{r_E}}.$$

Here Z denotes the control columns and $r_{EY \cdot Z}$ is the correlation of the two residual vectors. The numerator measures remaining linear co-variation; the denominator measures remaining education variation. This partial-regression identity gives the full model’s education coefficient.

Its correlation is unitless; multiplying by the residual-SD ratio expresses the slope in thousand dollars per education year.

Uncertainty must also account for the fitted controls. Prestige and women supply one column each, while type supplies two indicator columns. With education and the intercept, the full model therefore estimates $p = 6$ coefficients from $n = 98$ rows. Its residual degrees of freedom are $n - p = 92$. Using its residual sum of squares, RSS_{full} , gives

$$SE(b_E) = \sqrt{\frac{RSS_{\text{full}}/(n-p)}{\sum_i r_{E,i}^2}}, \quad t_E = r_{EY \cdot Z} \sqrt{\frac{n-p}{1-r_{EY \cdot Z}^2}}.$$

The slope identity is algebraic. The displayed SE estimates uncertainty under the independent, constant-variance error model; Normal errors give the exact t_{92} reference for $H_0 : \beta_E = 0$. Table 4 reads the result on the occupation records. The first row is the education row of the full model, which the residual formulas above reproduce; the second is a fresh fit of residual income on residual education.

Table 4

The residual slope reproduces the full-model coefficient; only the full model's df gives its uncertainty.

Calculation on the same 98 rows	Slope	SE	df	t	p
Full model <code>m2</code> , education row; the residual formulas with $n - p = 92$	0.1164	0.2752	92	0.4230	.6732
A fresh fit of residual income on residual education	0.1164	0.2694	96	0.4321	.6666

Both rows share the slope 0.1164, and the residual correlation is the partial correlation .0441. The fresh fit treats the residuals as ordinary data and counts $98 - 2 = 96$ df, as if fitting the four control columns had cost nothing, so its SE is smaller by the factor $\sqrt{92/96}$. Use `m2`, or the formulas above with $n - p = 92$, for uncertainty, and keep exactly the same rows in the auxiliary fits, the full fit and the correlations. The HC3 calculation in the next subsection retains the slope and uses a different covariance estimate.

This distinction also separates two meanings of covariance. Chapter 5's `cov()` describes how recorded variables spread and move together; the regression function `vcov()` estimates the covariance of the fitted coefficients. Table 5 compares them on the same 98 rows.

Table 5

Two covariance matrices from the same 98 records answer different questions.

	Recorded-data covariance	Coefficient covariance
R call	<code>cov()</code> of income, education, prestige and women	<code>vcov(m2)</code>
Rows and columns	4 numeric variables; the type factor is left out	6 fitted coefficients, including the intercept and two type indicators
Diagonal entries	Sample variances	Squared coefficient SEs
Off-diagonal entries	Sample covariances between variables	Joint uncertainty of two coefficients
Education entry	7.5567 years squared	0.0757; its square root is the SE 0.2752
Question it answers	How the recorded variables spread and move together across occupations	How uncertain the fitted coefficients are, and how their errors move together

The education entries differ by a factor of about 100 because they measure different things: the spread of education across occupations in years squared, and the squared uncertainty of one fitted coefficient in (thousand dollars per year) squared. The matrices are not interchangeable in a fitted-mean or contrast interval; Section 5 uses the coefficient covariance in $\mathbf{x}_*^T \widehat{\mathbf{V}} \mathbf{x}_*$.

Figure 2 sets the two comparisons side by side. The marginal line rises 0.8830 per year; after the fitted linear parts of prestige, women and type are removed from both variables, the residual line rises only 0.1164, and wide residual variation remains around it. Residualization does not remove all substantive differences between occupations. It removes the **fitted linear components** of the named variables. Chapter 9 develops the matrix version.

3.3 Read a coefficient interval under its stated model

The hypothetical linear conditional-mean model introduced in Section 1 has six estimated coefficients, so its residual degrees of freedom are $98 - 6 = 92$. For education the classical calculation is

$$b_E \pm t_{.975,92} SE(b_E) = 0.1164 \pm 1.9861(0.2752) \\ \approx [-0.4302, 0.6631].$$

The result is compatible, under those assumptions, with negative and positive conditional slopes on this scale. It does not demonstrate that education is unimportant, that the slope equals zero, or that adjustment has removed an education effect. No substantive equivalence margin was declared, and the interval includes differences of several hundred dollars per year of

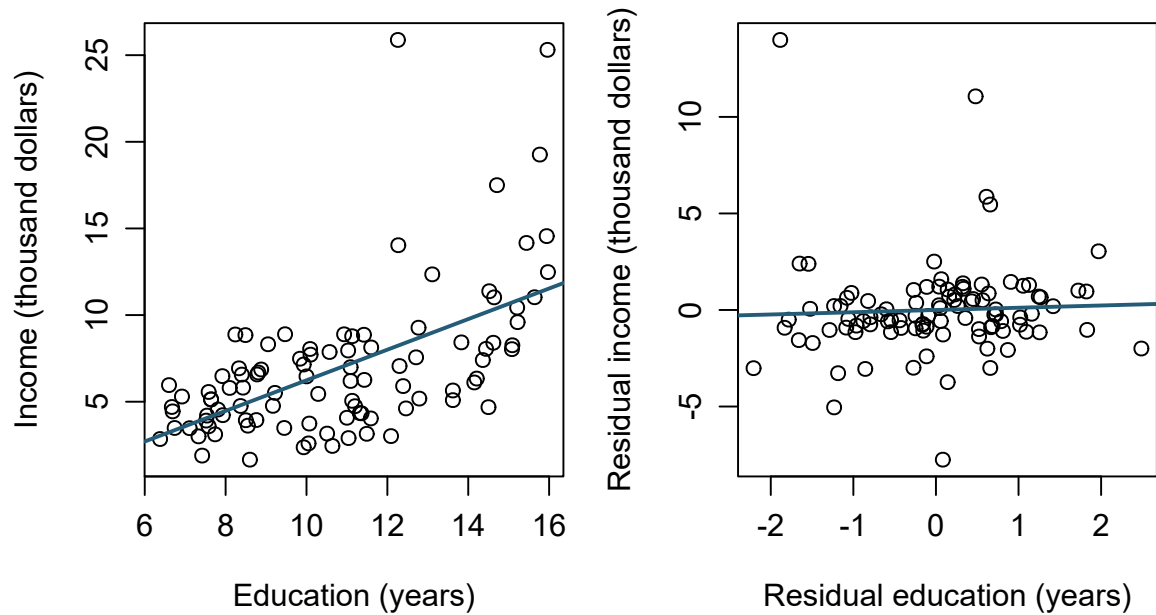


Figure 2: Marginal and residualized income–education comparisons in the same 98 occupations.

education. The data and model therefore leave the direction and practically relevant magnitude unresolved.

```
confint(m2, "education")
```

```

      2.5 %      97.5 %
education -0.4301895 0.6630567

```

```

hc3_covariance <- sandwich::vcovHC(m2, type = "HC3")
hc3_se <- sqrt(hc3_covariance["education", "education"])
coef(m2)["education"] + c(-1, 1) * qt(.975, 92) * hc3_se

```

```
[1] -0.6429122  0.8757794
```

HC3 gives SE 0.3823 and an approximate interval $[-0.6429, 0.8758]$ using the displayed t_{92} convention. It allows heteroskedasticity under independent-observation and other regularity conditions; it is not an exact small-sample guarantee. Both intervals leave broad uncertainty. Neither accounts for unmodeled dependence, selection, measurement error, or a wrong conditional-mean form.

If a test is reported, its target belongs with this inference: the two-sided education-coefficient test addresses $H_0 : \beta_E = 0$ in the stated full model, not “education has no effect.” Its classical $t = 0.423$ and $p = .6732$ do not prove the null. We do not use this test to choose which of the three questions deserves an answer.

4 Check Support, Form, Influence, and Precision

Checking asks what the model leaves unexplained and what follows for the answer. It is not a ritual in which each graph receives a pass or fail. Residuals, support, coefficient precision, and sensitivity each answer a different diagnostic question.

4.1 Read residual, scale, tail, and influence views together

A residual is an observed deviation from a fitted value, not the unobserved population error. A standardized residual uses a common residual-scale estimate; an externally studentized residual uses a scale estimate with that observation omitted. Leverage measures an unusual predictor configuration. Influence combines predictor and outcome information (Cook, 1977). The first extraction collects these quantities for every retained occupation and lists the five with the largest Cook’s distance.

```
ch8_diag <- data.frame(
  occupation = rownames(d), fitted = fitted(m2),
  residual = residuals(m2), standardized = rstandard(m2),
  studentized = rstudent(m2), leverage = hatvalues(m2),
  cooks = cooks.distance(m2)
)
ch8_top <- order(ch8_diag$cooks, decreasing = TRUE)[1:5]
ch8_diag[ch8_top, c("residual", "studentized", "leverage", "cooks")]
```

	residual	studentized	leverage	cooks
general.managers	14.206060	6.950201	0.07749926	0.44667800
physicians	11.009218	4.895873	0.07852799	0.27243082
ministers	-7.761571	-3.169572	0.03957245	0.06281250
lawyers	5.384776	2.166338	0.06284867	0.05043072
surveyors	-4.900673	-1.963544	0.06343196	0.04221084

The smooth and changing spread challenge a constant conditional-mean and variance description. The right tail matters to exact small-sample normal-theory inference. Neither feature prevents calculation of OLS. Large residual, high leverage, and high influence are different features; none is an automatic record-invalidity verdict. No residual graph can establish that occupations are independent draws from a wider population.

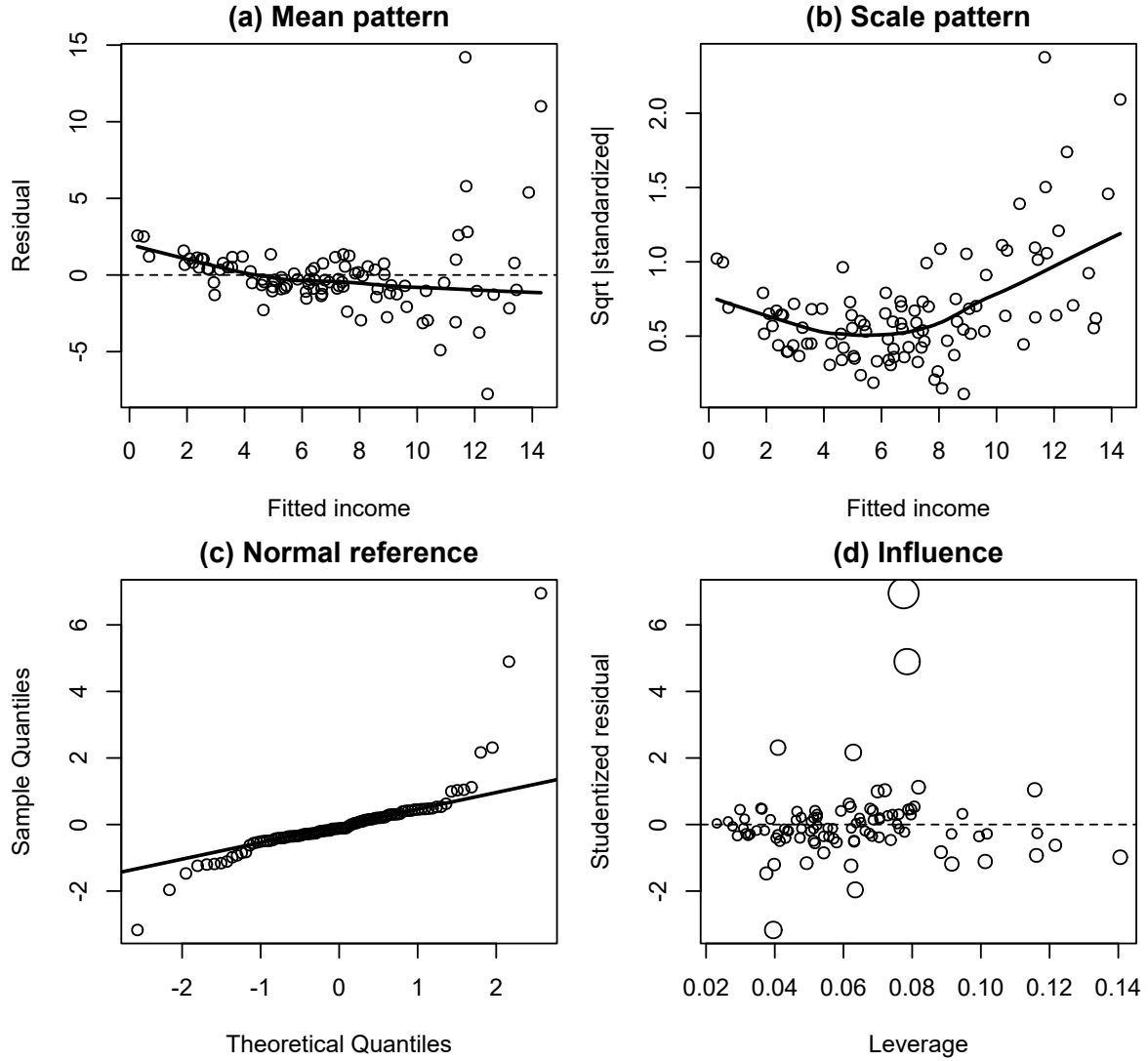


Figure 3: Four complementary diagnostic views for the full additive model.

4.2 Inspect joint support and coefficient precision

Joint support concerns predictor combinations, not merely separate minimum and maximum values. The convex-combination witness in Section 2 establishes support for the focal white-collar profile across all three continuous predictors. The marginal ranges below ask a different, coarser question: how far does each variable extend within each recorded type?

Table 6*Predictor ranges within each recorded occupation type.*

Type	n	Education	Prestige	Women
bc	44	6.38–10.93	17.3–54.9	0.00–90.67
prof	31	11.09–15.97	53.8–87.2	0.58–96.12
wc	23	9.17–12.79	26.5–67.5	3.16–97.51

Blue-collar education ends at 10.93 years, below the professional minimum of 11.09. A same-education comparison between those types necessarily crosses that gap. A model can return both predictions there; software output does not create observations. Lack of support limits what can be learned or interpreted about a defined comparison rather than making the estimand cease to exist.

The Computer Lab’s Round 1 reference, *Covariance objects, diagnostic displays and variations*, supplies the grid and plotting code used here. A conditional grid can also avoid pretending that an unrestricted rectangle is supported. For *m1*, hold prestige at 30, 50, or 70 and plot education only over values observed within 7.5 prestige points of each target. The width is an explicit display convention, not a formal support guarantee.

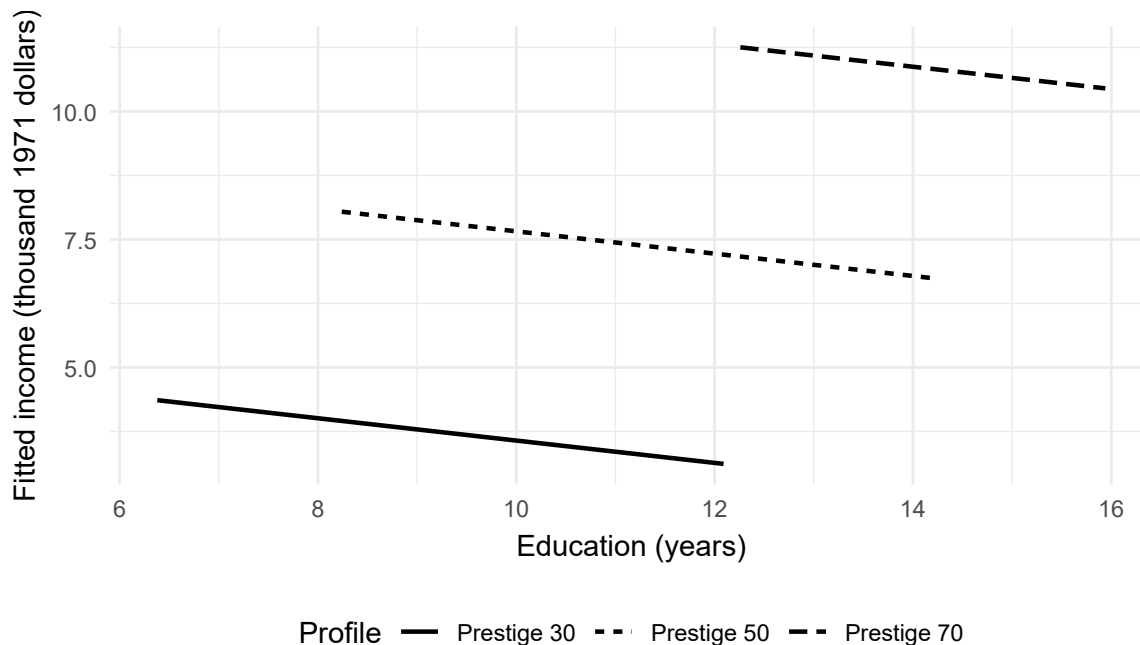


Figure 4: Neighborhood-restricted fitted profiles from the education-and-prestige model.

The parallel lines share *m1*’s education slope. This display conditions on prestige only, not on

the full set for Pair 2. It teaches a transparent plotting choice; it does not prove support for the full-model profile.

Education's variance inflation factor measures the collinearity cost. Regress education on the remaining predictors; the auxiliary R^2 measures how much of its variation they account for, and

$$VIF_E = \frac{1}{1 - R_{E|P,W,T}^2} \approx 8.0968, \quad \sqrt{VIF_E} \approx 2.8455,$$

since the auxiliary R^2 is .8765. Figure 5 draws the SE multiplier $\sqrt{VIF}$ against the auxiliary R^2 and marks the model's three continuous predictors. The curve rises slowly at first: women, whose auxiliary R^2 is .2312, loses little precision (multiplier 1.1405). Beyond about .8 it climbs steeply, and prestige (.8066, multiplier 2.2738) and education sit on that steep part.

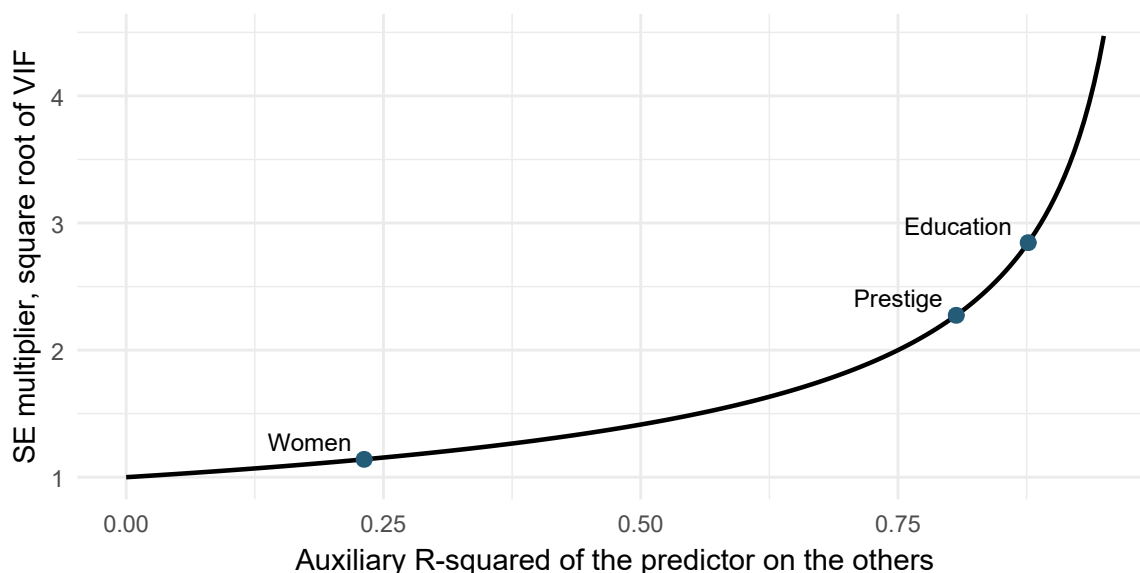


Figure 5: The SE multiplier grows steeply as the other predictors account for more of a predictor's variation.

Under the classical model, holding residual variance and total education variation fixed, education's coefficient SE is multiplied by 2.8455 relative to an orthogonal design. This is a comparison of variance formulas, not a claim that deleting other predictors will divide the actual SE by 2.8455. Deletion changes the comparison and can change residual variance. VIF is not an automatic deletion threshold, and near collinearity does not by itself create bias under the maintained zero-conditional-mean model.

4.3 Use targeted sensitivities to bound the answer

A component-plus-residual plot adds the fitted prestige contribution back to each residual, $e_i + \hat{\beta}_P P_i$. The smooth bends upward at high prestige relative to the straight fitted component. That pattern motivates a declared form sensitivity; it does not prove that one chosen curve is the true process.

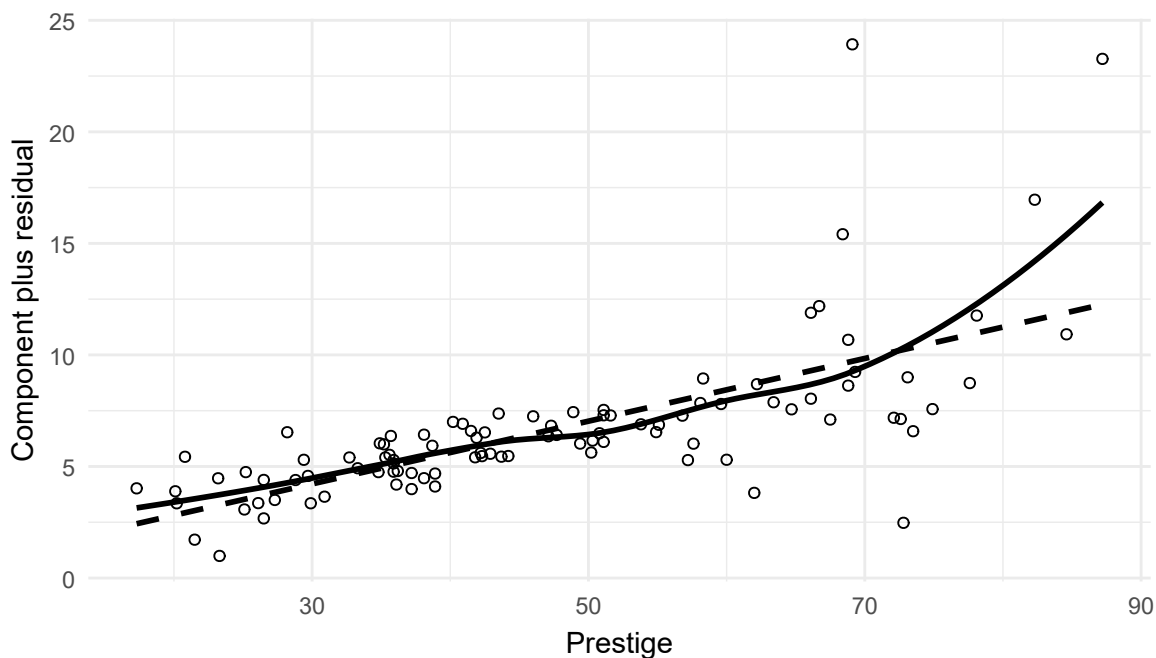


Figure 6: The prestige component plus residual in the full additive model.

The refit asks whether a squared prestige term improves on $m2$. The two fits are **nested**: they share the outcome and the 98 rows, and the smaller fit's predictors are a subset of the larger fit's, which adds one term. Chapter 6's nested-model F compared an additive and a full model of the feedback-and-practice study; the same comparison applies here, with continuous predictors. Its null hypothesis is that the squared-prestige coefficient is zero, so that $m2$'s mean function suffices. The statistic compares the residual sum of squares that the added term removes with the larger fit's residual variance, on 1 and $98 - 7 = 91$ degrees of freedom. A small p value says the added term removes more variation than the error scale would ordinarily produce under that null and the stated conditions. It does not establish that the enlarged equation is the correct mean function, and it is not a rule for selecting a model. Chapter 9 develops the general nested comparison.

```
ch8_curve <- update(m2, . ~ . + I(prestige^2))
anova(m2, ch8_curve)
```

Analysis of Variance Table

Model 1: income_k ~ education + prestige + women + type

Model 2: income_k ~ education + prestige + women + type + I(prestige^2)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	92	630.89				
2	91	587.41	1	43.481	6.736	0.01101 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
c(full_education = unname(coef(m2)["education"]),
  curve_education = unname(coef(ch8_curve)["education"]))
```

full_education	curve_education
0.11643361	-0.02980032

The classical nested comparison gives $F(1, 91) = 6.736$ and $p = .011$; education changes from 0.1164 to -0.0298 . In the enlarged equation, the prestige slope depends on prestige through $\beta_P + 2\beta_{P^2}P$. This is a focused sensitivity for Pair 2, not a validation-tuned replacement for the frozen prediction rule in Pair 3.

The record for general managers has the largest Cook's distance, about 0.4467. Its high income relative to the model is not proof of error. A transparent deletion refit, the same formula fitted to the other 97 records, asks how dependent the answer is on that record; the Computer Lab runs it. The education estimate moves from 0.1164 to 0.4372. Preserve the full-data estimate and disclose the sensitivity. Verify coding, measurement, and substantive distinctiveness before considering exclusion; a more attractive p value is not an exclusion rule. The refit is a diagnostic perturbation, not a new research question or evidence that the reduced sample is preferable.

Table 7

Checks change the bounded answer rather than producing one omnibus pass/fail score.

Check	What is visible here	Consequence for the answer
Functional form	Adding a prestige-squared term changes the education slope to about -0.0298	Treat the additive form as a working approximation; do not silently reinterpret it as an exact conditional law
Spread	HC3 raises the education SE from 0.2752 to 0.3823	Show sensitivity; a robust SE does not repair all model problems
Influence	Removing the record for general managers, which has the largest Cook’s distance, gives an education slope about 0.4372	Inspect the record and disclose sensitivity; do not delete it merely to obtain a preferred result
Collinearity	Education’s auxiliary R^2 is .8765	Little education variation remains after the other predictors; separate conditional coefficients can be imprecise
Joint support	Professional and blue-collar education ranges do not overlap	Cross-type comparisons at common covariates may depend heavily on extrapolation
Dependence and scope	Occupational aggregates have shared economic and classification contexts	Independence and same-source transport are assumptions, not verified by a residual plot

Appendix B develops Cook’s distance, coefficient covariance, generalized VIF, and model-comparison criteria, with a caution about comparing criteria across outcome scales; the Computer Lab carries the diagnostic and matrix reproductions. The results needed to interpret the answer are already in this section.

5 Move from Coefficients to Fitted and Standardized Quantities

The profile and standardization calculations in this section are **didactic extensions of the fitted model**, not extra question-facing answers owed by the occupation study. Pair 2 remains about β_E . A report that promoted one of these quantities to its own substantive conclusion would need to state the corresponding question, estimand, justification, and bounds.

5.1 A fitted mean is not prediction performance

Start with a complete input row and ask R for the two interval types. `interval = "confidence"` concerns the model mean at that profile; `interval = "prediction"` concerns one new outcome at the same profile. Both use the full fitted model and its stated classical assumptions.

```
profile <- data.frame(education = 11, prestige = 45, women = 50,
                      type = factor("wc", levels = levels(d$type)))
predict(m2, newdata = profile, interval = "confidence")
```

```
      fit      lwr      upr
1 5.588634 4.486469 6.690799
```

```
predict(m2, newdata = profile, interval = "prediction")
```

```
      fit      lwr      upr
1 5.588634 0.2722037 10.90506
```

Where do these intervals come from? For the white-collar profile in Section 2, collect the coefficient multipliers in $\mathbf{x}_* = (1, 11, 45, 50, 0, 1)^\top$. Their order is intercept, education, prestige, women, professional indicator, and white-collar indicator, and the superscript $\top$ denotes the transpose. With $\widehat{\mathbf{V}}$, the coefficient covariance of Section 3.2, the fitted mean and its estimated classical variance are

$$\widehat{m}_* = \mathbf{x}_*^\top \widehat{\boldsymbol{\beta}}, \quad \widehat{\text{Var}}(\widehat{m}_*) = \mathbf{x}_*^\top \widehat{\mathbf{V}} \mathbf{x}_*, \quad \widehat{\mathbf{V}} = \text{vcov}(m2).$$

The variance includes the coefficient variances **and their covariances**. For example, two terms $a\hat{\beta}_j + b\hat{\beta}_k$ contribute $a^2\widehat{\text{Var}}(\hat{\beta}_j) + b^2\widehat{\text{Var}}(\hat{\beta}_k) + 2ab\widehat{\text{Cov}}(\hat{\beta}_j, \hat{\beta}_k)$. Adding coefficient SEs, or ignoring covariance, is not the same calculation; take the square root only after calculating the variance. Here the fitted mean is 5.5886, its estimated variance 0.308 in squared units, and its SE 0.5549 thousand dollars, which gives the classical fitted-mean interval [4.4865, 6.6908] that `predict()` printed.

For **one new outcome**, its own error variation adds $s^2 = 6.8575$: the SE is $\sqrt{s^2 + \mathbf{x}_*^\top \widehat{\mathbf{V}} \mathbf{x}_*} \approx 2.6768$, giving the much wider interval [0.2722, 10.9051]. Figure 7 draws both intervals along the white-collar education range, holding prestige at 45 and women at 50. The dark fitted-mean band is narrowest near education 11.2, about 2.2 thousand dollars wide, and widens toward the edges of the records, to about 3.1 at 9.17 years. The light band for one new occupation stays between about 10.6 and 10.9 wide along the whole range, because the new occupation's error variance, 6.8575, dominates a fitted-mean variance of about 0.3.

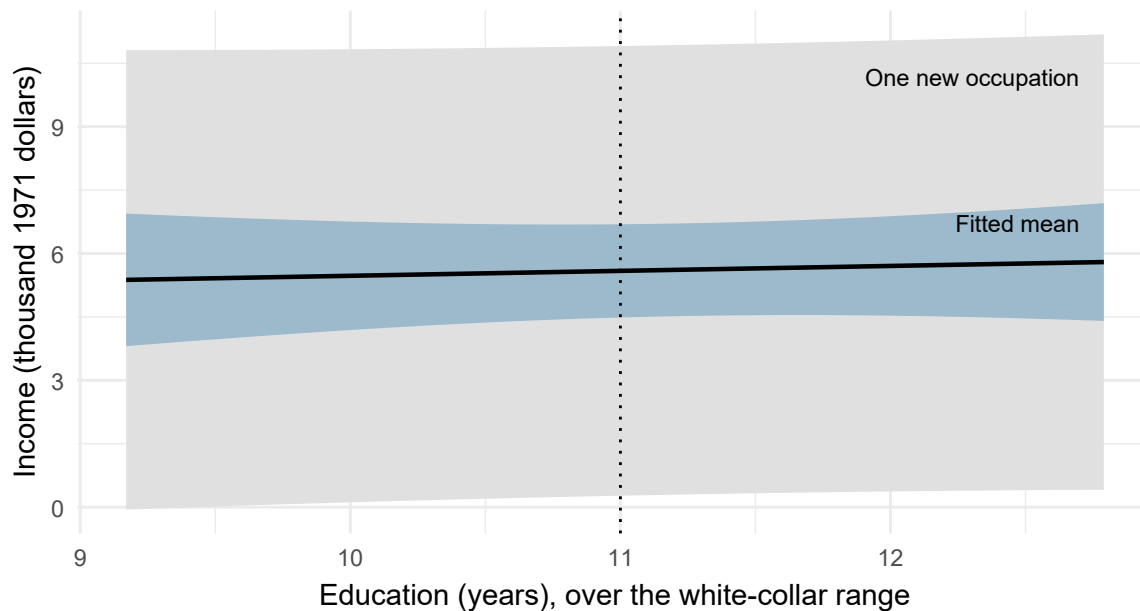


Figure 7: Two 95% intervals for a white-collar profile at prestige 45 and women 50, across the white-collar education range.

This calculation uses the independent normal common-variance conditional-mean model; it is not a promise established by the diagnostics. Replacing $\widehat{\mathbf{V}}$ with HC3 alone does not estimate an arbitrary new outcome's conditional error distribution. Neither interval tells us how accurately a learning rule predicts held-out records on average. That requires Pair 3.

5.2 Standardize over a named distribution

A conditional fitted value uses one specified profile. A standardized fitted value averages across a named distribution. To see the operation, copy the 98 observed profiles, change their type to white-collar, predict each fitted value, then take the mean. Each continuous profile has equal weight.

```
white_profiles <- d
white_profiles$type <- factor("wc", levels = levels(d$type))
mean(predict(m2, newdata = white_profiles))
```

[1] 7.016406

This is an average of model predictions after a specified profile change, not the observed white-collar group mean. For type g and fixed nonnegative weights w_i summing to one,

$$\hat{\mu}_g^{\text{std}} = \sum_i w_i \mathbf{x}_{ig}^T \hat{\boldsymbol{\beta}} = \mathbf{a}_g^T \hat{\boldsymbol{\beta}}, \quad \mathbf{a}_g = \sum_i w_i \mathbf{x}_{ig}.$$

Here $\mathbf{x}_{ig}$ is the design vector after assigning type g to observed continuous profile i . With empirical weights $w_i = 1/98$, this is standardization over these 98 profiles, not over a national population. Treating those profiles and weights as fixed gives estimated variance $\mathbf{a}_g^T \hat{\mathbf{V}} \mathbf{a}_g$.

For the white-collar quantity, averaging the six coefficient multipliers over the 98 changed profiles gives $\mathbf{a}_{\text{wc}}$, and $\mathbf{a}_{\text{wc}}^T \hat{\boldsymbol{\beta}}$ returns the same 7.0164. Its SE is 0.6177 and its classical interval [5.7896, 8.2432]; the variance includes the covariance between the predictions through their shared fitted coefficients, so averaging the 98 individual interval endpoints would give a different calculation. The Computer Lab computes $\mathbf{a}_{\text{wc}}$ and repeats the operation for each type. Table 8 sets the results beside the observed mean income of each type.

Table 8

Observed type means and empirical-profile standardization under the full additive model.

Type	Observed mean	Standardized estimate	SE	Lower	Upper
bc	5.3741	6.7811	0.6942	5.4023	8.1598
prof	10.5595	7.1053	0.9011	5.3156	8.8950
wc	5.0523	7.0164	0.6177	5.7896	8.2432

The observed means differ sharply: professional occupations average 10.5595 thousand dollars, about twice the other two types. They also have more education, higher prestige and different percentages of women. The standardized estimates give every type the same 98 profiles, so they differ only by the type coefficients, 0.3242 for professional and 0.2353 for white-collar relative to blue-collar.

Because $m2$ is linear and additive in the continuous predictors, its average fitted value equals its fitted value at their means. This is an algebraic special case: with squared terms, interactions, or a nonlinear response function, prediction at means need not equal mean prediction. Constant

additive type contrasts remain unchanged by common weights, although absolute standardized levels can change.

The overall education mean is 10.7951 years. Relabelling every profile professional or white-collar creates some unsupported combinations. Averaging them does not solve that problem. The white-collar profile used above is a different, locally supported conditional quantity; it is not a three-type comparison. The intervals condition on the empirical profiles and use the classical covariance. Estimating a new population's covariate distribution would add uncertainty not included here.

Standardization does not identify a causal effect by itself and does not validate out-of-sample performance. Pair 3 requires a learning rule and held-out outcomes.

6 Ask and Answer the Separate Prediction Question

6.1 Define the learning rule, new case, and loss

A learning rule says how training data become a fitted prediction equation. Our two rules are fixed: fit education-only OLS, or fit the full additive OLS equation with education, prestige, women, and type. Both use the same outcome and records. OLS **fits** by minimizing squared training errors; we separately choose **mean absolute error (MAE)** to evaluate the resulting predictions in interpretable income units.

A prediction opportunity here means a hypothetical same-source occupation with all declared predictors available before its income outcome is used. It does not mean predicting future earnings from information that would only become available afterward. The records do not establish that this hypothetical exchangeability assumption holds for new countries, periods, individual workers, or occupational types absent from the dictionary.

For held-out outcomes y_i and predictions $\hat{y}_i^{(-)}$ fitted without those outcomes,

$$MAE = \frac{1}{n_{\text{test}}} \sum_{i \in \text{test}} |y_i - \hat{y}_i^{(-)}|.$$

The superscript reminds us that the case was held out of fitting. A smaller number means smaller average absolute prediction error on the evaluated cases. It does not say that every case was predicted better.

6.2 Keep held-out outcomes out of training

We use shared ten-fold cross-validation. The 98 records are divided into ten folds; each fold takes a turn as the held-out set. Fit both equations on the other records, predict the held-out cases, then combine the 98 case-level errors.

Table 9

The internal-validation protocol was fixed before this teaching comparison.

Decision	Frozen teaching protocol	Why it matters
Candidates	Education-only and the declared full additive OLS rule	No search for the best-looking model after validation
Folds	Shared balanced random ten-fold assignment, seed 80806	Both rules face the same held-out cases
Training sizes	Eight fits use 88 records; two use 89	Risk concerns those training sizes, not a model trained on all 98
Preparation	Same 98 complete records; fixed income/1,000 conversion and type dictionary	No outcome-informed preprocessing; selection still limits scope
Estimability	Every training set must contain all three types and have full column rank	Stop rather than silently substitute a different rule
Summary	Pool absolute errors over all 98 held-out cases	Each case receives equal weight; folds of different size do not
Sensitivity	200 new resplits of the same records, seed 80807	Describes sensitivity to splitting, not new independent studies

This is a retrospective teaching reanalysis of familiar historical data. “Fixed before this comparison” is not a claim of historical preregistration or that analysts had never seen the outcomes. The candidates are not revised after inspecting the primary errors; the association sensitivities in Section 4 are not covert additions to the validated candidate set.

```

RNGkind("Mersenne-Twister", "Inversion", "Rejection")
set.seed(80806)
fold <- sample(rep(seq_len(10), length.out = nrow(d)))
held_fold <- fold[1]
train <- d[fold != held_fold, ]
held <- d[fold == held_fold, ]
c(fold = held_fold, training = nrow(train), held_out = nrow(held))

```

```

      fold training held_out
      9         89         9

```

```

fold_simple <- lm(income_k ~ education, data = train)
fold_full <- lm(income_k ~ education + prestige + women + type,
               data = train)
first_case <- d[1, ]
first_predictions <- c(
  Education = unname(predict(fold_simple, first_case)),
  Full = unname(predict(fold_full, first_case)))
first_predictions

```

```

Education      Full
 8.94613  11.59482

```

```
abs(first_predictions - first_case$income_k)
```

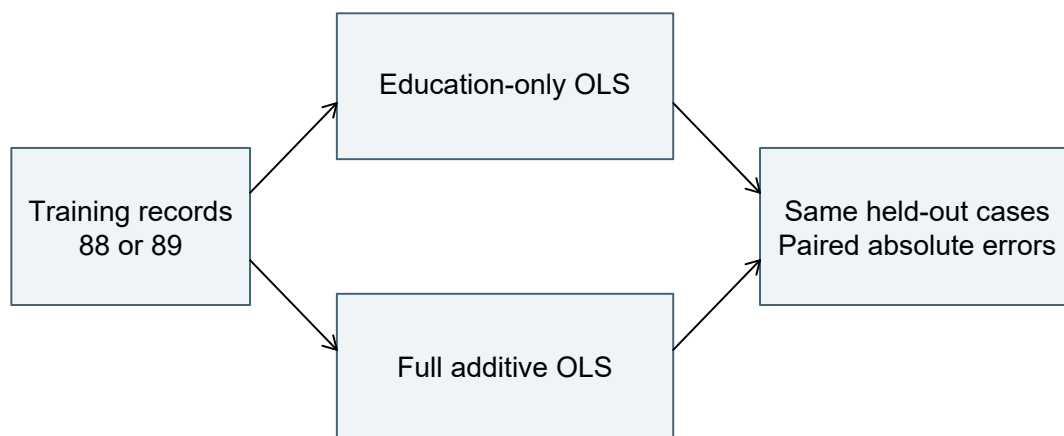
```

Education      Full
3.4048703 0.7561808

```

This is one fold of the frozen protocol. Government administrators, the first source record, falls in fold 9, which leaves 89 training records and 9 held-out records; the record was excluded from both fits, and the code predicts it. Every training set must contain all three types for the full rule to be estimable; the Computer Lab checks this and the fitted ranks for every fold. Repeating the same train/predict/error steps across all ten folds gives the all-98 comparison below. One record's error is not that summary.

The Computer Lab provides the complete ten-fold function and the 200-resplit code; Chapter 13 develops held-out validation fully. The visible one-fold example needs only the preceding core data definitions, not that helper. No held-out outcome is used to estimate a coefficient. If we introduced learned scaling, imputation, or variable selection, each step would have to be fitted using training data only. A fixed unit conversion such as dollars to thousands is not an estimated preprocessing step.



Repeat across ten folds; pool the 98 case-level errors.

Figure 8: Shared held-out cases let two fixed learning rules face the same prediction task.

6.3 Work one error, then read the comparison

For government administrators, the first source record, observed income is 12.351 thousand dollars. Its held-out fold is 9. The full-rule prediction is 11.5948, giving

$$|12.351 - 11.5948| = 0.7562.$$

The education-only prediction is 8.9461, giving $|12.351 - 8.9461| = 3.4049$. This one record favors the full rule. We still need all 98 errors:

Table 10

Case-weighted absolute errors from the frozen shared-fold comparison.

Learning rule	Held-out MAE (thousand dollars)
Education-only OLS	2.4031
Full additive OLS	1.5412
Difference: full minus education-only	−0.8619

The full rule’s average held-out absolute error is about 862 dollars lower in this split. Because we subtract the two errors for **each same case** before averaging, difficulty differences between validation cases do not come from using different test sets for the two rules. A different loss can

be read from the same held-out predictions: the root mean squared error, which weights large errors more heavily, is 3.5051 for education-only and 2.7111 for the full rule, so the preference does not change here; it need not agree with MAE in other data.

What is the performance estimand? Eight test folds contain 10 cases whose predictions were trained on 88; two contain 9 cases trained on 89. Consequently the pooled quantity targets the declared mixture

$$R_{\text{mix}} = \frac{80}{98}R_{88} + \frac{18}{98}R_{89},$$

where R_m is expected absolute error of the specified learning rule trained on m eligible same-source records and used on a new one under the stated repetition. Pair 3 compares the two rules' R_{mix} . It does not directly estimate performance of the single equation fitted to all 98 observed records.

The 200 resplits give full-minus-education MAE differences from -0.914 to -0.741 , all favoring the full rule. This supports stability to the examined split choices. It is **not a 95% interval**: the same cases are reused, training sets overlap, and the results are not independent studies. We do not test a mean of ten fold errors as if those errors were independent. Uncertainty across newly drawn training samples is not quantified here. The estimated risk advantage is therefore provisional; its sign is not established for the hypothetical source population by the resplit range.

The comparison supports lower internal held-out error for the full rule in these records and under this protocol. It does not isolate the predictive contribution of education: the rules differ by prestige, women, and type together. Nor does it establish external transport, individual earnings prediction, or a causal education effect.

7 Read the Coordinated Arguments, Then Evaluate and Synthesize

7.1 The chapter-authored study record

The following report consolidates the preceding analysis. Treat its paragraphs, tables, equations, and code as an accessible teaching record, not as statements copied from the source investigators.

R0 – Common record and declared scope. We analyzed the 98 complete occupation records described in Section 1. Income was scaled to thousand 1971 Canadian dollars. Education and income were historical occupational averages; prestige came from an earlier rating survey. We retained the same rows for all three questions and make no claim of verified probability sampling or randomized education.

R1 – Marginal answer. Across these records the education-only least-squares slope was 0.883 thousand dollars per year. This is a descriptive linear summary: occupations with more education tended to have higher average income. It is not an individual or causal effect.

R2 – Conditional answer. The intended estimand was the linear conditional-mean education coefficient β_E , comparing education at the same prestige, percentage of women, and occupational type. Its fitted estimate was $b_E = 0.1164$, with a classical 95% interval $[-0.4302, 0.6631]$ under the independent linear conditional-mean working model with common variance. HC3 gave $[-0.6429, 0.8758]$. Influence, functional-form, and support checks limit that model’s interpretation. The direction and practically relevant magnitude remain unresolved under these calculations; the result does not establish no conditional relationship.

R3 – Prediction answer. The two fixed OLS learning rules were compared with the shared ten-fold protocol in Section 6. The full rule had MAE 1.5412 versus 2.4031 for education-only, difference -0.8619 . The examined resplits all favored the full rule. This is internal evidence about learning procedures trained on 88/89 eligible records under the stated same-source exchangeability assumptions. Lower expected error is a provisional conclusion because new-sample uncertainty is not quantified; this is not external validation of a final 98-record equation.

R4 – Synthesis. Education has a positive marginal linear association with income in these historical records. Its conditional association is imprecisely resolved by the specified additive model, while the full learning rule predicts the held-out records more accurately than the education-only rule. These conclusions answer different questions and do not establish that changing education would change income.

7.2 Read the functions and their traces

A component is a logical function; the record contains study-specific content serving that function. A passage can carry several functions. For example, R0 identifies records for STAT-E and supplies historical context for SUB-E. It becomes J content only when it offers a reason connecting those records to a particular inference. “We used R” does not by itself supply that reason.

Table 11

Reading the linked-layer functions without three duplicate maps.

Function	Shared or pair-specific content	Trace in this teaching record
STAT-E	Named 98 rows, variables, exclusions, and computations’ inputs	R0; the row audit and import code in Section 1

Function	Shared or pair-specific content	Trace in this teaching record
SUB-E	Occupations rather than workers; historical dollars, education, ratings, and types	R0; variable/source descriptions
STAT-J	Pair 1: fixed-record least-squares definition. Pair 2: declared stochastic model and uncertainty conditions. Pair 3: held-out fitting, fixed rules, loss, and exchangeability	R1–R3; model declaration; validation protocol and code
SUB-J	Reasons connecting those measurements and comparisons to historical occupational income, with limits on individual, current, and causal interpretations	R0–R3; unit/time explanations and prediction-opportunity definition
STAT-I	Pair 1’s 0.8830 slope; Pair 2’s 0.1164 estimate, intervals, coefficient test and checks; Pair 3’s case-level errors, MAEs, and split sensitivity	Section 3 through Section 6; R1–R3
STAT-C	Descriptive positive marginal slope; unresolved conditional slope under the model; smaller internal error for the full rule under the protocol	R1, R2, and R3 respectively
SUB-I	Interpret years, income units, conditioning, and prediction errors in the named historical occupation setting	Worked substitutions and error calculation; R1–R3
SUB-C	Each question-facing answer with its own limits; none supplies a causal or individual-earnings answer	R1–R3; their distinctions preserved in R4

SUB-E situates evidence; SUB-J **augments and contextualizes** statistical justification; SUB-I interprets the inference; SUB-C bounds the substantive answer. Sharing a dataset does not make Pair 2’s independent-error model into Pair 3’s validation procedure.

For each supporting statement, record what the report says and where the reader can find it: the report itself, the data documentation, or the executable calculation record. State gaps and conflicts in ordinary prose. For example, R2 states the independence assumption, but this teaching record does not supply evidence establishing independence under the original occupation-selection process. A stated assumption and evidence for it are different pieces of information.

Similarly, the income-unit conversion makes the interpretation of 0.883 thousand dollars as about 883 dollars traceable without inventing an empirical premise. No conflicting source is

identified in this defined record; a real study may contain incompatible descriptions that a reader should record and investigate. Here, a reason’s documentary status means what has been stated or located, and which support is missing. It is not a quality score or a four-category classification to learn.

Professional knowledge helps identify what function a trace serves. It does not authorize adding an unreported design feature. If an author gave the invalid reason “all rows are in the same dataset, therefore they are independent,” it would still be **stated J**, whose adequacy is judged separately. Reading records the claim actually offered, not a repaired claim the reader wishes had been offered.

7.3 Evaluate each argument separately

We now change from identifying the reported reasoning to evaluating its quality. The six Chapter 1 questions apply to each argument. They are not six scores to total, and not every concern needs to be counted repeatedly.

Table 12

Apply the six questions before evaluating the study-level synthesis.

Quality-evaluation question	Application to these arguments
Question alignment	R1 matches its finite-record slope. R2 names β_E , its fitted estimate, the conditioning set, and the model extension. R3 names the learning procedure, loss, and training sizes rather than substituting a coefficient test.
Evidence integrity	The 102-to-98 accounting and common rows are explicit. Missing type still limits what is represented. CV excludes held-out outcomes from fitting; source selection is not repaired by that procedure.
Justification adequacy	Least squares is sufficient for the declared finite description. Pair 2’s model conditions and Pair 3’s same-source exchangeability remain assumptions with incomplete empirical support. Statistical computation alone cannot establish them.
Inferential adequacy	The slope, intervals, error arithmetic, and comparisons are reproducible. Neither zero-crossing nor a large p-value proves absence; repeated folds are not independent replication. Substantive reasoning must retain those distinctions.
Transition fidelity	Occupation-level average income remains occupation-level average income; partial adjustment is not a causal intervention, and model fit is not held-out performance.

Quality-evaluation question	Application to these arguments
Claim calibration	R1 remains finite and historical; R2 reports unresolved magnitude/direction rather than no relationship; R3 remains internal and procedure-specific. No claim extends to individual workers or modern populations.

At an individual question, useful judgments include **adequate for this purpose**, **needs revision**, **insufficient information to judge**, or **not applicable**, with a reason. For example, the record is adequate to check Pair 3's held-out computations but gives insufficient information to establish exchangeability for an external population. These judgments assess adequacy; they are separate from recording what the report says and where its support can be found.

Each whole pair then receives its own conclusion:

- **Pair 1: supported as stated.** The answer is an accurate bounded description of the retained records.
- **Pair 2: supported as stated**, as the explicitly conditional, unresolved working-model answer about β_E actually reported. This does not certify the model or establish a population relationship. A claim that the true national conditional slope is zero would not be supported by this record.
- **Pair 3: supported as stated**, as a qualified internal comparison with assumptions and limits retained. External superiority of the final all-record fitted equation is **not assessable from the accessible record**; it is not the reported answer being accepted.

The other available whole-pair conclusions are **supported only after narrowing or revision**, **not supported**, and **not assessable from the accessible record**. A correctly reported inconclusive answer can be supported as stated. Evaluating what the study actually says prevents a bounded answer from being rejected merely because it does not answer a stronger question.

7.4 Add Cross-question coherence for the synthesis

For coordinated questions, ask one additional question: **Does the synthesis preserve the distinct questions, estimands, methods, analytic statuses, uncertainty, and limits of the separate answers?** This is **Cross-question coherence**, introduced locally here. Supplementary B develops it further but is not needed to apply it.

R4 is coherent because it reports three distinct outcomes. A positive marginal slope, an unresolved conditional slope, and better internal prediction can coexist. They do not contradict one another.

Now consider the alternative sentence:

Once the other predictors were controlled, education had no effect; nevertheless, the significant full regression proves it will predict workers' incomes accurately.

Reading first identifies the offered inference and claim. Evaluation then finds several failures: “no effect” substitutes a causal null for the conditional-association answer; an omnibus test is not validation; and occupations become workers. No omnibus significance test can replace the held-out errors or rescue the change of unit. The corrected synthesis is R4, not a fourth overarching argument that erases the separate verdicts.

7.5 Auxiliary construction callback

The construction workflow supports researchers carrying out the study; it is not a mandatory reading sequence. Each question retains its own six-stage cycle from Chapter 1. For coordinated questions, Supplementary B's extension aligns the plans at Stage 3 and adds Stage 7 for synthesis.

In **Understand and plan**, researchers review the problem and literature, frame questions and candidate answers, and align design, measurement, estimands, evidence acquisition, and inference. Here, “separate questions” means separately owed conclusions. If the obligation were one integrated conclusion, jointly necessary estimands would be planned as strands instead.

The extended **Stage 3, Plan and align prospectively across questions**, specifies common rows and units but distinct marginal, model-based, and validation plans. In **Produce and complete, Stage 4, Produce or acquire and document evidence**, designates the question-specific records; **Stage 5, Estimate, check, and analyse**, performs the computations; and **Stage 6, Contextualize, interpret, and complete the bounded answer**, finishes each answer with its actual limits. **Stage 7, Give explicit outcomes and synthesize**, records all outcomes and brings them together without creating an omnibus argument.

Iteration remains within a phase. A same-anchor error correction or justified uncertainty repair can reopen Phase 2 work. A newly preferred outcome, question mode, or estimand requires a new or revised cycle, with reused records where appropriate and its timing disclosed. A newly chosen causal question cannot be “narrowed” into association and treated as answered. This teaching reanalysis does not retrospectively claim that the historical data were collected for our three questions.

8 Reading Bridge

Multiple regression adds conditional comparisons, not automatic causal control. Subtracting fitted means showed what a coefficient compares; residualization connected it to covariance; residual, influence, support, and precision checks had different consequences for a bounded answer; a fitted profile and a standardized quantity were didactic extensions of the conditional model; and held-out errors answered a separately owed question. The occupation study owed three answers: a finite-record marginal description, a working-model conditional interval with unresolved magnitude, and a qualified internal prediction comparison. Shared records made the questions coordinated without making their justifications identical. The next regression question is different: might the education comparison itself vary with another predictor rather than remain the same additive slope? Chapter 9 first collects the tools this chapter used only briefly: factors with several levels and their contrasts, nested model comparison and information criteria, changed predictor and outcome forms, and robust, clustered, and bootstrap uncertainty, and it transfers the reading discipline to Project STAR, where a randomized design supplies a different warrant. Chapter 10 then extends the interaction reasoning from Chapters 4 and 6 to these regression comparisons.

Key terms

Use this short reference to revisit the concepts introduced here. The Part One Concept Reference gives the broader reference and the principal home of each concept; the simple-regression, interval, ANOVA, and reading vocabulary used in this chapter is explained in Chapters 1–4 and 7.

Table 13

Selected concepts for multiple regression across coordinated questions.

Term and common alternatives	Meaning
Multiple regression (additive model)	A fitted equation in which the outcome's mean is the sum of an intercept and one term per predictor. Each coefficient compares fitted means at the same values of the other predictors; the additive form assumes each comparison is the same across those values.
Partial slope (conditional coefficient; adjusted coefficient)	The difference in fitted outcome per one-unit difference in one predictor, holding the model's other predictors fixed. Its meaning depends on the conditioning set named with it; it is not an intervention effect or a claim about unmeasured differences.

Term and common alternatives	Meaning
Indicator variable (dummy variable) and reference category	A 0/1 variable marking membership in one category of a factor; a factor with J levels needs $J - 1$ indicators, and the omitted level is the reference. The intercept refers to the reference category; each indicator coefficient is a difference from it at the same other predictors. Releveling changes coordinates, not fitted values.
Partial regression (residualization; added-variable form)	Regressing both the outcome and one predictor on the other predictors, then relating their residuals. The residual slope equals the partial slope; the residual correlation is the partial correlation, unit-free.
Coefficient covariance matrix	The estimated variances and covariances of the fitted coefficients (<code>vcov()</code> in R). Any linear combination of coefficients, such as a fitted mean or a contrast, has variance $\mathbf{c}^T \widehat{\mathbf{V}} \mathbf{c}$; adding SEs or ignoring covariances is a different, incorrect calculation.
Variance inflation factor (VIF)	One divided by one minus the R^2 from regressing a predictor on the other predictors. Its square root multiplies that coefficient's SE relative to an orthogonal design, holding residual variance fixed; it describes precision, not bias, and is not a deletion rule.
Joint support (covariate hull)	Whether a combination of predictor values lies within the region where observations actually occur, not merely within each variable's separate range. A fitted value outside joint support is extrapolation even if each coordinate is observed somewhere.
Standardization over a named distribution (model-averaged fitted value)	Averaging the model's fitted values after setting one predictor to a chosen value for every profile in a declared set with fixed weights. Its uncertainty uses the coefficient covariance; it is not an observed group mean, a causal effect, or held-out performance.
Learning rule (learning procedure)	A fixed recipe that turns training records into a prediction equation, such as fitting a declared OLS formula. Prediction performance concerns the rule at stated training sizes, not one final fitted equation.
Cross-validation (held-out validation)	Dividing records into folds, fitting a rule on the other folds, predicting the held-out fold, and pooling the errors so that no outcome contributes to its own prediction. Resplits describe sensitivity to the partition, not new independent studies.
Mean absolute error (MAE)	The average absolute difference between held-out outcomes and their predictions, in outcome units. It is an evaluation loss chosen separately from the squared-error criterion that OLS minimizes.
Cross-question coherence	The additional evaluation question for coordinated answers: does the synthesis preserve the distinct questions, estimands, methods, uncertainty, and limits of the separate answers rather than merging them into one omnibus claim?

Further reading

- The OpenIntro multiple-predictor tutorial offers concrete equations and indicator interpretations. We use that equation-first teaching device without adopting automatic stepwise selection as this chapter’s route.
- The ISLR regression lab connects fitting, coefficients, checks, and predictions (James et al., 2021).
- The ISLR cross-validation lab develops the train/predict/check sequence (James et al., 2021). Our fixed rules, MAE, and training-size qualifications are specified for this teaching example.
- The carData documentation (Fox et al., 2022; Fox & Weisberg, 2019) supplies the Prestige variable meanings and provenance. The coordinated study and R0–R4 report are chapter-authored, not claims attributed to that documentation.
- Fox (Fox, 2016) and Kutner and colleagues (Kutner et al., 2005) develop the linear-model algebra behind partial slopes, indicators, and diagnostics.

Supplementary B develops coordination and its auxiliary construction workflow in greater detail.

Exercises

Use the supplied numbers; rounding within 0.02 thousand dollars is acceptable unless a different precision is requested. These prompts do not require Supplementary B or the appendices.

Exercise 1: Keep the three answers distinct (Foundational)

A report promises: (a) a description of education and income across 98 occupations, (b) an education comparison at the same prestige, women, and type, and (c) a comparison of two prediction rules on held-out records.

1. Explain why these are three question–answer pairs rather than three statistical strands serving one integrated answer.
2. For each, name its estimand in plain language and one kind of output that could contribute to its answer.
3. Does sharing the same 98 records establish the same justification for all three? Give one concrete reason.

Exercise 2: Name what changes (Foundational)

Classify each change as primarily changing the records, the comparison, the coefficient coordinates, or the uncertainty calculation. State whether it necessarily creates a new scientific question.

1. Refit education-only regression after excluding four missing-type records.
2. Add prestige, women, and type to the education-only association equation.
3. Change the reference category from blue-collar to white-collar.
4. Keep the fitted equation but replace classical SEs with HC3 SEs.

Finally, explain why appropriately specified adjustment in a randomized study need not change its declared average assignment-effect estimand.

Exercise 3: Interpret a fitted comparison (Foundational)

Use this rounded hypothetical equation, with income in thousand dollars:

$$\hat{y} = 1 + 0.20e + 0.10p - 0.04w + 0.30D_{\text{prof}} + 0.15D_{\text{wc}}.$$

1. Two white-collar profiles share prestige $p = 40$ and women $w = 50$; one has education $e = 12$ and the other $e = 10$. Which terms cancel when their fitted means are subtracted? Give the modeled difference and interpret it in dollars.
2. What does the coefficient 0.15 compare, and at which values of the other predictors?
3. What must be checked before calling the education comparison in part 1 supported by the observed data?
4. Why is the modeled difference not an estimate of the effect of giving a worker two more years of education?

Exercise 4: Read held-out errors (Foundational)

Four held-out occupations have the observed incomes and predictions below, in thousand dollars. Each prediction was fitted without the corresponding outcome.

Occupation	Observed	Education-only prediction	Full-rule prediction
1	5	4	6
2	8	10	8
3	10	7	9
4	13	12	11

1. Calculate the MAE of each rule.
2. Explain why the average of the four paired differences, full minus education-only, equals the difference between the two MAEs, and why it matters that both rules predict the same four occupations.
3. Does the better average rule have to be better for every occupation? Point to the table.
4. Explain why this comparison is neither a coefficient interval nor validation of a model subsequently fitted to all available records.

Exercise 5: Locate a reason before judging it (Integrative)

An accessible report states:

We retained the same 98 occupations for both equations. This makes the observations independent. The full equation has higher training R^2 , so we conclude it predicts new occupations more accurately.

The report provides import code and fitted-model output, but no validation split, held-out predictions, or study-specific evidence about dependence.

1. Identify the reasons offered, the inference, and the claim.
2. Record the documentary status of the independence reason and the held-out prediction evidence separately. Do not rewrite “stated” as “not located” merely because a reason is wrong.
3. Apply Justification adequacy, Inferential adequacy, and Claim calibration.
4. Give a whole-pair evaluation conclusion for the prediction answer and name the additional work it would require.

Exercise 6: Evaluate the study-level synthesis (Integrative)

Use the occupation study's reported results: marginal slope 0.883; fitted conditional estimate $b_E = 0.116$ of β_E , with classical interval $[-0.430, 0.663]$; and held-out MAEs 2.403 for education-only versus 1.541 for the full rule.

A summary says:

Education does not matter after adjustment, but the model proves that increasing workers' education raises income and predicts earnings well in other countries.

1. Read the claims actually offered before evaluating them.
2. Identify which separate answer obligation is being replaced or exceeded by each clause.
3. Apply Cross-question coherence and write a corrected three-sentence synthesis that retains the three reported answers and their limits.
4. If the authors now want a causal effect of education, is that a repair of their existing uncertainty calculation or a new/revised question-level cycle? Explain.

Appendix A: Adjustment, Coding, and Group-Comparison Mathematics

This appendix develops the mathematics behind Section 2. Appendix B retains technical reproduction and sensitivity; Chapter 9 develops contrasts, block tests, adjusted means, the matrix form, and the Project STAR transfer. The occupation calculations use the same 98-row object d and the models $m0$, $m1$, and $m2$ defined in the main text. They do not introduce another sample. The type codes remain *bc*, *prof*, and *wc*.

Why the marginal slope includes another association

Suppose a population process is

$$Y = \beta_0 + \beta_X X + \beta_Z Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0,$$

with finite variances and $\text{Var}(X) > 0$. The marginal least-squares slope is a covariance divided by a variance. Substituting the outcome equation gives

$$\begin{aligned} \frac{\text{Cov}(X, Y)}{\text{Var}(X)} &= \frac{\beta_X \text{Var}(X) + \beta_Z \text{Cov}(X, Z) + \text{Cov}(X, \varepsilon)}{\text{Var}(X)} \\ &= \beta_X + \beta_Z \frac{\text{Cov}(X, Z)}{\text{Var}(X)}. \end{aligned} \tag{1}$$

The zero-error-covariance condition removes the final term. The discrepancy therefore involves both the association between X and Z and Z 's coefficient in the outcome model. Neither ingredient alone determines its size. If the target is the conditional coefficient, omission creates bias relative to that target. If the target is the marginal slope, that marginal quantity is not a biased version of an unstated conditional target. Neither interpretation supplies a causal warrant for observational variables.

Changing a specification does not universally change a scientific estimand. Different estimators can target the same declared quantity, as Chapter 9's randomized adjustment example, Project STAR, illustrates. Releveling a factor merely changes coordinates. By contrast, the marginal and conditional coefficients in this example name different comparisons.

A known process and repeated samples

Let independent standard normal variables X and U generate $Z = .65X + \sqrt{1 - .65^2}U$. Then both predictors have variance one and their correlation is .65. Independently generate a normal error with SD 6, and set

$$Y = 50 + 2X + 5Z + \epsilon. \quad (2)$$

This is the process of Section 2: the marginal slope is $2 + 5(.65) = 5.25$ and the conditional X slope is 2. The sample drawn in Figure 1 has 250 observations and uses seed 82108. Its marginal slope is 5.9971 and its conditional slope 3.0259: near their targets but not on them, as one sample's estimates should be. The Computer Lab's Round 1 reference, *Covariance objects, diagnostic displays and variations*, supplies the complete generator and fitted models.

Now use a different, explicitly stated sample size: 150 observations in each of 1,000 repetitions. This is a distinct repeated-study design, with seed 82109; it is not repetition of the 250-observation draw. The same Computer Lab reference supplies the full simulation and density-plot code.

Table 14

Targets and results across 1,000 generated samples.

Model	Target	Mean	SD
Marginal	5.25	5.2581	0.5952
Conditional	2.00	2.0010	0.6707

Both procedures center near their own targets. The conditional estimates are more variable here: separating correlated predictors uses less residual predictor variation. This is a result of the specified process, not a general promise that adjustment always lowers precision. Prognostic

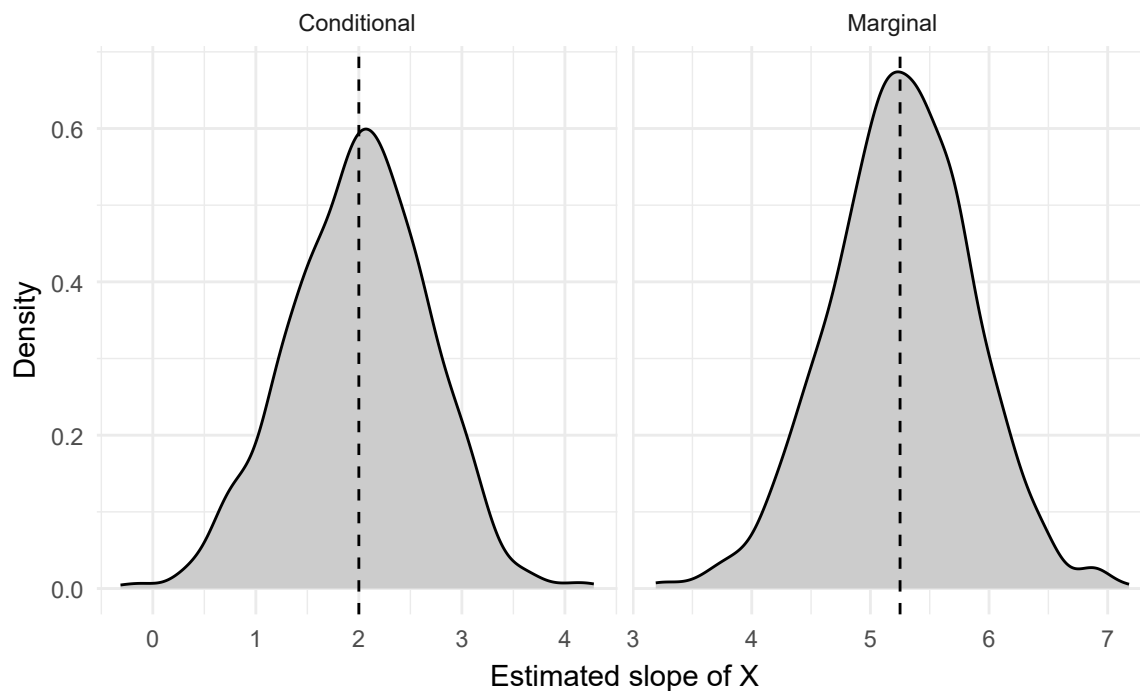


Figure 9: Sampling distributions around two different targets.

covariates and their relations to the focal predictor both matter. Simulation demonstrates consequences of its assumptions; it cannot certify the observational occupation study.

Indicators recover fitted group means

Start with a model containing only the three-level type factor. With blue-collar as reference,

$$\hat{y}_g = \hat{\alpha} + \hat{\delta}_{\text{prof}} D_{\text{prof}} + \hat{\delta}_{\text{wc}} D_{\text{wc}}.$$

The fitted means are $\hat{\mu}_{\text{bc}} = \hat{\alpha}$, $\hat{\mu}_{\text{prof}} = \hat{\alpha} + \hat{\delta}_{\text{prof}}$, and $\hat{\mu}_{\text{wc}} = \hat{\alpha} + \hat{\delta}_{\text{wc}}$. OLS minimizes squared deviations separately within each group, so these equal the observed group means. The hats distinguish fitted summaries from hypothetical population means.

Table 15 reads the fit on the 98 records. The group-only coefficients are $\hat{\alpha} = 5.3741$, $\hat{\delta}_{\text{prof}} = 5.1853$ and $\hat{\delta}_{\text{wc}} = -0.3218$; each fitted mean combines them as the indicator columns direct, and each equals the observed mean income of its type. Sums of the rounded coefficients can differ from the fitted means in the last digit.

Table 15

The type-only regression reproduces the three observed group means.

Type	D_{prof}	D_{wc}	Coefficients combined	Fitted mean	Observed mean
bc	0	0	$\hat{\alpha}$	5.3741	5.3741
prof	1	0	$\hat{\alpha} + \hat{\delta}_{\text{prof}}$	10.5595	10.5595
wc	0	1	$\hat{\alpha} + \hat{\delta}_{\text{wc}}$	5.0523	5.0523

This group-only description is not the adjusted type comparison in *m2*. In *m2*, indicators compare fitted values at the same education, prestige, and percentage women. The factor is one term represented by two columns; it is neither an ordered numeric score nor one selected reference-category coefficient.

Releveling changes the intercept and indicator coefficients, not the fitted values, residuals, or a fixed substantive contrast.

The same Round 1 reference in the Computer Lab reproduces this releveling operation and checks the unchanged fitted values with complete code and output.

Releveling reproduces the same fit up to floating-point noise. No new question has appeared merely because the coordinates changed. Centering continuous predictors also changes the intercept's reference profile without changing fitted values. It cannot create cross-type support.

The overall F test

The overall regression test sets all $p - 1$ non-intercept coefficients to zero. Under the same normal-error conditions as the coefficient tests, its statistic is

$$F_{\text{overall}} = \frac{\text{RegSS}/(p - 1)}{\text{RSS}/(n - p)}, \quad \text{RegSS} = \sum_i (\hat{y}_i - \bar{y})^2.$$

```
summary(m2)$fstatistic
```

```
value  numdf  dendif
32.1747  5.0000 92.0000
```

The full model gives $F(5, 92) = 32.1747$; the education-only model on the same 98 records gives $F(1, 96) = 47.1956$, which the Computer Lab prints beside it. Each tests its stated joint restriction. Neither verifies the mean function or shows how well the fitted rule will predict a held-out outcome. Chapter 9 develops the general nested comparison behind this statistic, its matrix form, and the contrasts, block tests, and adjusted group means that a factor with several levels invites.

Appendix B: Technical Reproduction and Sensitivity

Sections 4 and 5 carry the ordinary support, diagnostic, standardization, and targeted-sensitivity lessons. This appendix supplies mathematical detail and worked results; the Computer Lab supplies the longer computational reproductions. It uses d , $m0$, $m1$, and $m2$ without changing their source rows. Classical intervals are conditional-model benchmarks; technical refinements do not create missing design, measurement, independence, or transport warrants.

Cook's distance and full deletion accounting

The main route interprets residual, leverage, and influence views and shows the education-slope sensitivity. This section supplies the corresponding Cook's distance formula and complete coefficient accounting.

With p coefficient columns including the intercept, leverage is $h_{ii} = \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i$. Cook's distance can be calculated as

$$D_i = \frac{e_i^2}{ps^2} \frac{h_{ii}}{(1 - h_{ii})^2} = \frac{(\hat{\beta} - \hat{\beta}_{(-i)})^\top \mathbf{X}^\top \mathbf{X} (\hat{\beta} - \hat{\beta}_{(-i)})}{ps^2}.$$

The second expression measures a coefficient change using the design matrix's metric, not an unweighted sum of squared coefficients with incompatible units.

Table 16

The five largest Cook's distances in the full fit.

Occupation	Residual	Leverage	Cook's distance
general.managers	14.2061	0.0775	0.4467
physicians	11.0092	0.0785	0.2724
ministers	−7.7616	0.0396	0.0628
lawyers	5.3848	0.0628	0.0504
surveyors	−4.9007	0.0634	0.0422

The record for general managers has the largest Cook's distance, approximately 0.4467. The main route explains why this observation is inspected and retained.

Refitting without that record moves education from 0.1164 to 0.4372 (Section 4), and the type coordinates move as well; the Computer Lab prints the full coefficient comparison. That comparison records a diagnostic perturbation of the fit, not six new research questions or evidence that the reduced sample is preferable.

HC3 changes covariance, not the records or target

For independent observations, the HC3 covariance estimator is

$$\widehat{\mathbf{V}}_{\text{HC3}} = (\mathbf{X}^\top \mathbf{X})^{-1} \left\{ \sum_i \mathbf{x}_i \mathbf{x}_i^\top \frac{e_i^2}{(1 - h_{ii})^2} \right\} (\mathbf{X}^\top \mathbf{X})^{-1}.$$

The outer factors are the familiar design component. The middle component allows different residual scales and adjusts for leverage (Long & Ervin, 2000; Zeileis, 2004). It leaves the OLS coefficients unchanged.

The Computer Lab’s Round 1 reference, *Covariance objects, diagnostic displays and variations*, supplies the manual matrix reproduction of this expression and compares it with `vcovHC()`. The first classical/HC3 comparison in Section 3.3 already supplies the education interval; the table gives the full coefficient context.

Table 17

Coefficient estimates and two covariance summaries.

Term	Estimate	Classical SE	HC3 SE
(Intercept)	0.4205	1.9573	2.7741
education	0.1164	0.2752	0.3823
prestige	0.1406	0.0354	0.0387
women	−0.0535	0.0097	0.0079
typeprof	0.3242	1.4631	2.0928
typewc	0.2353	0.9848	1.0548

Education’s HC3 SE is 0.3823 rather than 0.2752. Using t_{92} as an approximate reference gives $[-0.6429, 0.8758]$. This is not an exact finite-sample HC3 coverage result. Women’s HC3 SE is smaller than its classical counterpart: robust covariance need not widen every interval.

For a maintained linear conditional mean, HC3 addresses unequal variance under independent observations. For a random-design best-linear-projection target, a suitable sandwich covariance can support large-sample projection inference without making the mean exactly linear. Neither interpretation repairs measurement error, dependence, selection, causal identification, or absent joint support.

VIF describes coefficient precision

For a single predictor column, classical conditional variance can be written

$$\text{Var}(\hat{\beta}_j | \mathbf{X}) = \frac{\sigma^2}{\sum_i (X_{ij} - \bar{X}_j)^2} \frac{1}{1 - R_j^2}.$$

Here R_j^2 comes from regressing that predictor on the remaining columns, including the intercept. The variance inflation factor is $VIF_j = 1/(1 - R_j^2)$; its square root is an SE multiplier, holding the residual variance and focal predictor variation fixed (Kutner et al., 2005).

Section 4.2 reads education's VIF from Figure 5. The Computer Lab computes it from the auxiliary regression and with `car::vif(m2)`, which also reports the two-column type term.

Education gives $R_j^2 = .8765$, VIF 8.0968, and SE multiplier 2.8455. The predictor correlation between education and prestige is .8665, but VIF uses all the other columns, not just that pair.

The two-column type term has a generalized VIF. Its raw GVIF is 7.7001; the dimension-adjusted value $GVIF^{1/(2df)} = 1.6658$ is on a comparable scale to the square root of a one-column VIF (Fox & Weisberg, 2019). Do not compare a raw two-column GVIF with an ordinary one-column VIF as if their dimensions matched.

Exact collinearity creates rank deficiency. Near collinearity leaves coefficients estimable but potentially imprecise. Under the maintained zero-conditional-mean model it does not itself bias OLS. A cutoff of 5 or 10 neither establishes all assumptions nor justifies removing a variable required by the question. Measurement problems also need their own evidence: the familiar simple attenuation story requires classical error assumptions and is not a universal rule for every coefficient in a multiple-predictor model.

A local Normal likelihood and in-sample comparison criteria

Chapter 6 varied the candidate probability for a fixed binomial count of 32 successes in 50 trials. Here the data remain fixed again, but the response is continuous and the parameters are several regression coefficients plus an error scale. For a declared independent Normal linear model, the **likelihood** is the joint density of the observed outcomes at their fixed predictor values, viewed as a function of those parameters. It is not a probability assigned to a parameter. A continuous density value also differs from the probability of an exact observed vector. The **log-likelihood** adds log densities across the independent records.

Write the predictor row for record i as $\mathbf{x}_i^T$ and let $\sigma > 0$ be the error SD. Before fitting, the log-likelihood is

$$\ell(\boldsymbol{\beta}, \sigma) = -n \log \sigma - \frac{n}{2} \log(2\pi) - \frac{1}{2\sigma^2} \sum_i (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2.$$

At any fixed positive scale, only the last term changes with the coefficients. Maximizing it over $\boldsymbol{\beta}$ therefore minimizes the residual sum of squares: the Normal maximum-likelihood coefficients equal the OLS coefficients. This connects Chapter 7’s squared-loss criterion to this particular distributional model. Least squares itself did not require that model to calculate a minimizing line.

Maximizing over scale gives $\hat{\sigma}_{\text{ML}} = \sqrt{RSS/n}$. The residual SD used for classical coefficient uncertainty is $s = \sqrt{RSS/(n-p)}$, with p counting the regression coefficients. The two denominators serve different estimation purposes; substitute the ML scale when checking the fitted likelihood. At the fitted coefficients and ML scale,

$$\ell(\hat{\boldsymbol{\beta}}, \hat{\sigma}_{\text{ML}}) = -\frac{n}{2} \{ \log(2\pi) + 1 + \log(RSS/n) \}.$$

R’s `logLik()` returns this maximized value for an `lm()` fit by default, and adding the Normal log densities of the 98 outcomes at their fitted means with the ML scale returns the same number; the Computer Lab carries that check. Two information criteria combine this maximized log-likelihood with a parameter-count penalty:

$$AIC = -2 \log \hat{L} + 2k, \quad BIC = -2 \log \hat{L} + k \log n.$$

Here k counts the regression coefficients **and** the estimated residual variance. For the full occupation model, six coefficient columns plus that variance give $k = 7$. Smaller values are preferred under each criterion; neither is observed held-out MAE (Akaike, 1974; Schwarz, 1978).

The ML scale is 2.5373 and the residual SD is 2.6187. The maximized log-likelihood is -230.302 , so the full model’s AIC is 474.6040 and its BIC 492.6988. Table 18 compares the three declared models using the same 98 rows, response scale, and likelihood convention. The Computer Lab’s Round 2 reference, *All three types, one weighting change and the criteria*, supplies the criterion-table construction. Chapter 11 develops likelihood fitting further for binomial outcomes and numerical calculations; that later treatment is not needed to read this local Normal-model calculation.

Table 18*In-sample summaries for comparable income models.*

Model	k	R^2	Adjusted R^2	AIC	BIC
m0	3	.3296	.3226	526.5052	534.2601
m1	4	.4996	.4891	499.8328	510.1727
m2	7	.6362	.6164	474.6040	492.6988

The raw AIC/BIC values for income and log income must not be compared as if they used the same response density. A scale change changes the likelihood's reference measure; a valid common-scale comparison would need the corresponding transformation adjustment. The simpler lesson here is to preserve the outcome and question in each comparison.

No residual graph establishes independence. The occupations share one historical labour market, and metadata do not establish a probability sample of a wider population. For students within schools or repeated observations, the design may require dependence-aware uncertainty. HC3 alone is not such a repair.

The practical response is consequential rather than pass/fail: retain the named comparison, report its form and influence sensitivities, distinguish classical from robust uncertainty, and bound interpretation by measurement, support, and source. Causal adjustment-set rules about confounders, mediators, and colliders apply to a declared causal target; they are not universal requirements for association or prediction.

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